

If AlexNet aimed to improve performance by "deepening" the network, GoogLeNet (Inception v1) took a different approach: while "widening" the network, it carefully designed the architecture to control computational cost.

Its biggest highlight is the introduction of a completely new basic module- the inception module.

The ingenious use of 1x1 convolution (bottleneck layer): This is the key to keeping GoogLeNet lightweight. Before the expensive 3x3 and 5x5 convolutions, a 1x1 convolution is used to reduce the number of channels of the input features (dimensionality reduction), thereby significantly reducing the computational load while increasing the non-linearity of the network.

Parallel pooling: The module also includes pooling operations, providing another path for feature extraction for the network, and uses 1x1 convolution to adjust the number of channels, facilitating the final concatenation.

Concatenation is the final step of the Inception module. The outputs of the four branches are stacked together along the channel direction while maintaining the same height and width (both 28x28).

